Reconstructing the orientation of compact-binary mass posteriors from the mass-ratio marginal

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Abstract

The two-dimensional component-mass posterior of a compact-binary gravitational-wave event is elongated along a contour of near-constant chirp mass, a feature usually treated as a nuisance degeneracy. We show that its *orientation* is quantitatively reconstructible from a single one-dimensional marginal of the same posterior. Mapping the event's mass-ratio marginal through the constant-chirp-mass curve predicts the principal-axis position angle to a median 1.26° on GWTC-4.0/O4a and 1.22° on GWTC-5.0/O4b for elongated posteriors (axis ratio ≥ 3), with no coefficient calibrated on either catalog; the prediction was fixed on GWTC-3 and registered before the later data were examined. The corresponding local tangent approximation, which underlies rapid parameter-estimation tools, achieves 4.20–6.67° on the same events. The result does not follow from a curve merely fitting a curved distribution: substituting any other event's mass-ratio marginal degrades the error by 3.2–11.3×, and the achieved error lies below the minimum of 300 catalog-stratified permutations in all three catalogs. Inputs taken from one waveform family recover a different family's independently sampled axis to 2.08°–2.78°, against a 1.99° scale set by the families' mutual disagreement, so the result is not an artifact of reusing one posterior's Monte Carlo noise. The residual $\sim 1°$ is 17.2 times the Monte Carlo resolution of the released samples and is therefore a property of the model rather than of finite sampling; it is tracked by the curve's failure of principal-curve self-consistency ($\rho = 0.68$, $p = 7 \times 10^{-12}$), and finite posterior thickness is an out-of-sample contributor. We present this as a posterior-compression relation and a systematics diagnostic. It is not a test of general relativity: the posteriors are generated with general-relativistic waveform models, and their internal geometry cannot bound departures from the theory that produced them.

1 Introduction

Parameter estimation for a compact-binary coalescence yields a posterior probability distribution over the source parameters. In the component-mass plane this distribution is characteristically elongated: the chirp mass

$$\mathcal{M}_c = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}} \tag{1}$$

dominates the leading-order inspiral phase and is measured far more precisely than the mass ratio, so the posterior extends along a contour of near-constant $\mathcal{M}_c$ [1; 2; 4]. This shape is routinely described qualitatively and treated as a degeneracy to be marginalised over.

We ask whether the shape carries quantitative, predictable structure. Specifically: can the orientation of the (m_1, m_2) posterior — the position angle of its principal axis — be recovered event by event from low-dimensional summaries of the same posterior, using no coefficient calibrated on the data being predicted, and does such a reconstruction survive on later, disjoint catalogs?

It can. Mapping an event's mass-ratio marginal through the constant-$\mathcal{M}_c$ curve predicts the measured position angle to about a degree on elongated posteriors, in two catalogs released after the prediction was fixed. The organising idea is that the uncertainty geometry of gravitational-wave inference is largely *measurement optics* — a property of what the signal and instrument can resolve, rather than of source astrophysics — and that this geometry is precise enough to be used as a diagnostic.

The ingredients are individually familiar. That inference problems possess a few stiff and many sloppy directions is the content of the sloppy-model framework [7]; that the principal axis of data spread along a curved arc differs from the local tangent is the distinction between linear principal component analysis and a *principal curve* [8]. The published work closest to ours stops short in identifiable ways. Hannam et al. [4] state that "the component masses consistent with the signal lie roughly along a line of constant chirp mass" and label chirp mass on their confidence regions, but make no quantitative statement about orientation. Ohme et al. [5] apply principal component analysis to post-Newtonian phase coefficients rather than to (m_1, m_2) posterior samples, predict no per-event orientation, and perform no catalog validation. The `simple-pe` metric of Fairhurst et al. [6] assumes the match "varies quadratically with the difference in parameters" about a peak, yielding an error ellipse — the same local, linear approximation as our tangent baseline, though computed in $(\mathcal{M}, \eta)$ rather than (m_1, m_2) coordinates — and those authors note that they "observe curved contours rather than perfect ellipses". We are not aware of published work that reconstructs per-event posterior orientation from the chirp-mass curve over an event's own mass-ratio marginal, or that validates such a reconstruction out-of-sample across catalogs.

Section 2 defines the construction and the data. Section 3 reports the out-of-sample performance and the tests that establish it is not trivial. Section 4 examines the origin of the residual, and Section 5 its robustness to coordinate choice and to the elongation threshold. Section 6 discusses scope and limitations.

2 Method and data

2.1 The predicted orientation

For an event with source-frame posterior samples $\{(m_1, m_2)\}$, let ψ_{meas} be the position angle of the principal axis of the sample covariance and let the axis ratio axr be the ratio of its principal standard deviations. The constant-$\mathcal{M}_c$ curve in the mass plane is

$$m_1(q) = \mathcal{M}_c\,(1+q)^{1/5} q^{-3/5}, \qquad m_2 = q\,m_1, \tag{2}$$

with $q = m_2/m_1 \in (0, 1]$ and $\mathcal{M}_c$ the event's median source-frame chirp mass. The predicted orientation ψ_{curve} is the principal axis of the point set obtained by evaluating Eq. (2) over the event's own mass-ratio marginal. The local linear baseline ψ_{tangent} is the tangent to the constant-$\mathcal{M}_c$ contour at the posterior median point. No coefficient is calibrated on the validation catalogs; every ingredient is supplied by the event itself.

2.2 The prediction depends only on the mass-ratio marginal

The construction has one input, not two. Changing $\mathcal{M}_c$ multiplies $m_1(q)$ and $m_2(q)$ by a common factor, which is a dilation about the origin; a dilation leaves the eigenvectors of the sample covariance unchanged. The predicted orientation is therefore *exactly* invariant to $\mathcal{M}_c$, and ψ_{curve} is a functional of the mass-ratio marginal alone.

The chirp mass remains necessary to place the curve in the mass plane, which the thickness and self-consistency analyses of Section 4 and the geometry shown in Figure 3 require, but it contributes nothing to the orientation. One consequence is used in Section 5.1: repeating the analysis in detector-frame masses changes the *measured* axis only, never the prediction.

This is a compression statement rather than a prediction from external information. The mass-ratio marginal is drawn from the same posterior whose orientation is reconstructed, so what is demonstrated is that a two-dimensional posterior's orientation is recoverable from one of its own one-dimensional marginals, using no information from the (m_1, m_2) covariance. Section 3.2 quantifies how much of the accuracy that marginal supplies.

2.3 Data and event selection

We use public parameter-estimation samples from three releases: GWTC-2.1 [14] and GWTC-3 [15], which together form the derivation set; GWTC-4.0/O4a [9]; and GWTC-5.0/O4b [10]. The two later catalogs are mutually disjoint and disjoint from the derivation set, sharing no events. Samples are required to be finite in all mass parameters and to satisfy $0.02 < q \leq 1$.

The analysis is restricted to *elongated* posteriors, axr ≥ 3. This is a physical requirement rather than a tuned threshold: as a covariance approaches isotropy its principal axis ceases to be defined, and any orientation prediction must degrade toward a random baseline. Section 5.2 shows that the score varies smoothly across the threshold and that nothing distinguishes the chosen value.

The relation was fixed on the derivation set and registered before any later-catalog file was opened; the scores in Table 1 are the resulting locked decision. Appendix C describes the computational pipeline and the provenance of every number reported here.

3 Results

3.1 Out-of-sample performance

On elongated events the median mismatch $|\psi_{\text{curve}} - \psi_{\text{meas}}|$ is 1.26° on O4a and 1.22° on O4b. Table 1 collects the locked scores together with the tangent baseline and the replication statistics; Figure 1 shows the per-event comparison behind those medians, drawing each elongated event as a line joining its tangent error to its curve error so that the full distribution is visible rather than a summary.

3.2 The event's own mass-ratio marginal is required

That a curve should describe a curved distribution better than a straight line is unsurprising. The substantive question is whether the *event's own* marginal carries the information, or whether any plausible mass-ratio distribution would perform comparably. It does not (Figure 2).

Replacing the event's marginal with one pooled over the catalog, or with another event's, degrades the median error by 3.2–11.3×. Under 300 catalog-stratified permutations of the mass-ratio marginals among events, the achieved error lies below the *minimum* of the null distribution in every catalog, giving a permutation p-value $< 1/300$ throughout. We report the full permutation

	GWTC-3 (derivation)	GWTC-4.0/O4a	GWTC-5.0/O4b
median $\lvert\psi_{\rm curve} - \psi_{\rm meas}\rvert$ (elongated)	—	**1.26°**	**1.22°**
median $\lvert\psi_{\rm curve} - \psi_{\rm meas}\rvert$ (all)	—	7.53°	5.41°
median $\lvert\psi_{\rm tangent} - \psi_{\rm meas}\rvert$ (all)	7.49°	9.36°	8.23°
chirp-vs-total win fraction	78%†	86%	88%
Spearman($\lvert\Delta\psi_{\rm tangent}\rvert$, axr)	−0.42	−0.40	−0.69
N events (elongated)	75	86 (19)	104 (32)

Table 1: Locked out-of-sample scores, computed on the full released posterior samples. The reconstruction lands within $\sim$1.2° of the measured orientation on elongated events in two later, disjoint catalogs. The figures recompute these values independently from the cached extract (Appendix C) and agree: 1.26° on O4a against the locked 1.26°, and 1.19° on O4b against 1.22°, the latter differing only in which waveform group is taken as preferred. The tangent error *decreases* with elongation (final row), i.e. orientation becomes better defined for more elongated posteriors; normalised arc length behaves the same way, correlating negatively with the tangent error in all three catalogs ($\rho = -0.51, -0.57, -0.49$). †No locked win fraction exists for the derivation catalog, so that cell is computed from the cached extract; the other two are locked.

distribution rather than a single shuffled assignment because one draw is not a null: in O4a the 300 draws span 3.19–10.35°, so any individual shuffle could have fallen anywhere in that range.

3.3 Transfer between waveform families

Because the mass-ratio marginal is taken from the posterior being reconstructed, a natural concern is circularity: the agreement might reflect shared Monte Carlo noise rather than shared geometry. We test this by crossing waveform families. Using family A's mass-ratio marginal to predict family B's independently sampled principal axis — never touching B's own geometry — gives 2.08° (A→B) and 2.78° (B→A) over 64 events elongated in both, against tangent baselines of 3.57° and 6.25° in the same directions. The reference scale is 1.99°, the median angle between the two families' own measured axes.

The reconstruction therefore transfers between separately sampled posteriors of the same event and is not an artifact of reusing one posterior's Monte Carlo fluctuations. The test has a definite limit: the two families analyse the same strain data with shared calibration and priors, so it excludes reuse of sampling noise but not modelling error common to both. We describe 1.99° as a reference scale rather than an error floor, since a reconstruction may in principle denoise a posterior functional and fall below the raw disagreement between families.

4 Origin of the residual

4.1 The residual exceeds the Monte Carlo resolution

The $\sim 1°$ residual is not sampling noise. Bootstrapping the released samples — drawing one index set and recomputing *both* the measured axis and the prediction inputs from it, so that their correlated Monte Carlo errors cancel — gives a median resolution of 0.07° on the difference, against a median discrepancy of 1.07°: a ratio of 17.2. The constant-$\mathcal{M}_c$ curve is an accurate but imperfect description of posterior orientation, and the gap is a property of the model rather than of the sample count.

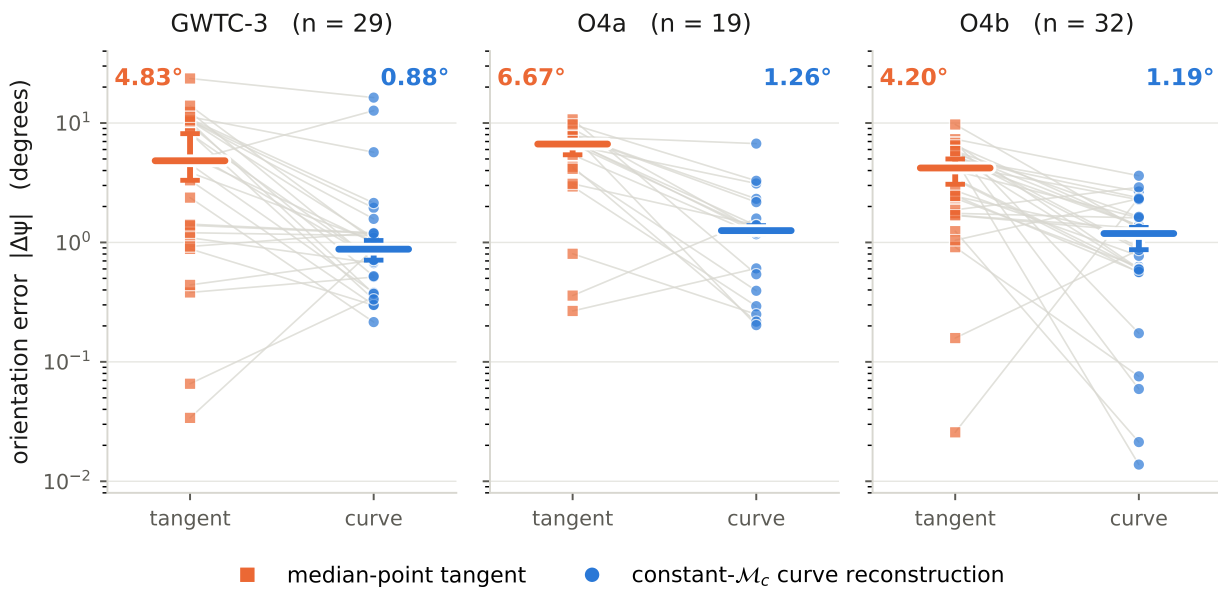

Figure 1: Reconstruction versus the local linear approximation, on elongated events (axr ≥ 3) in three disjoint event catalogs. Each thin line is one event, joining its median-point tangent error to its constant-$\mathcal{M}_c$ curve reconstruction error; heavy bars are catalog medians with a bootstrap interval computed over events. Medians: GWTC-3 ($n = 29$): $4.83° \to 0.88°$, O4a ($n = 19$): $6.67° \to 1.26°$, O4b ($n = 32$): $4.20° \to 1.19°$. The vertical scale is logarithmic because per-event errors span roughly 0.01–24°. The interval is a bootstrap over the event sample, not the per-event Monte Carlo resolution of Sec. 4.

This bootstrap measures the Monte Carlo resolution of the released sample set. It shrinks as more samples are released and encodes nothing about detector noise, calibration, waveform systematics or population variation; it is not a coverage statement, and we do not report the fraction of events falling within one or two bootstrap widths as though it were.

The residual carries a signed component: the curve under-rotates relative to the measured axis by a median 0.75°. This is not uniform across catalogs, being significant in the derivation catalog ($p < 0.001$) and in O4a ($p = 0.019$) but not in O4b ($p = 0.110$), the newest and largest of the three. We therefore report it per catalog.

4.2 Failure of principal-curve self-consistency

Hastie & Stuetzle [8] define a principal curve by *self-consistency*: $f(t) = \mathbb{E}[X \mid t_f(X) = t]$, so that every point of the curve is the mean of the data projecting onto it. This is directly testable for the constant-$\mathcal{M}_c$ curve, and it fails. Projecting each posterior sample onto its nearest curve point and comparing that point with the mean of the samples landing there leaves a perpendicular offset of about 2.6% of the posterior's own scale: the constant-$\mathcal{M}_c$ curve is not a principal curve of the posterior.

The magnitude of that failure tracks the reconstruction residual. The per-event self-consistency violation correlates with the per-event orientation residual at Spearman $\rho = 0.68$ ($p = 7 \times 10^{-12}$) on the primary projection grid, with ρ between 0.68 and 0.69 across a sweep of grid resolutions, so the result does not depend on that choice. Both quantities are measured rather than fitted, so the correlation cannot be inflated by additional model freedom.

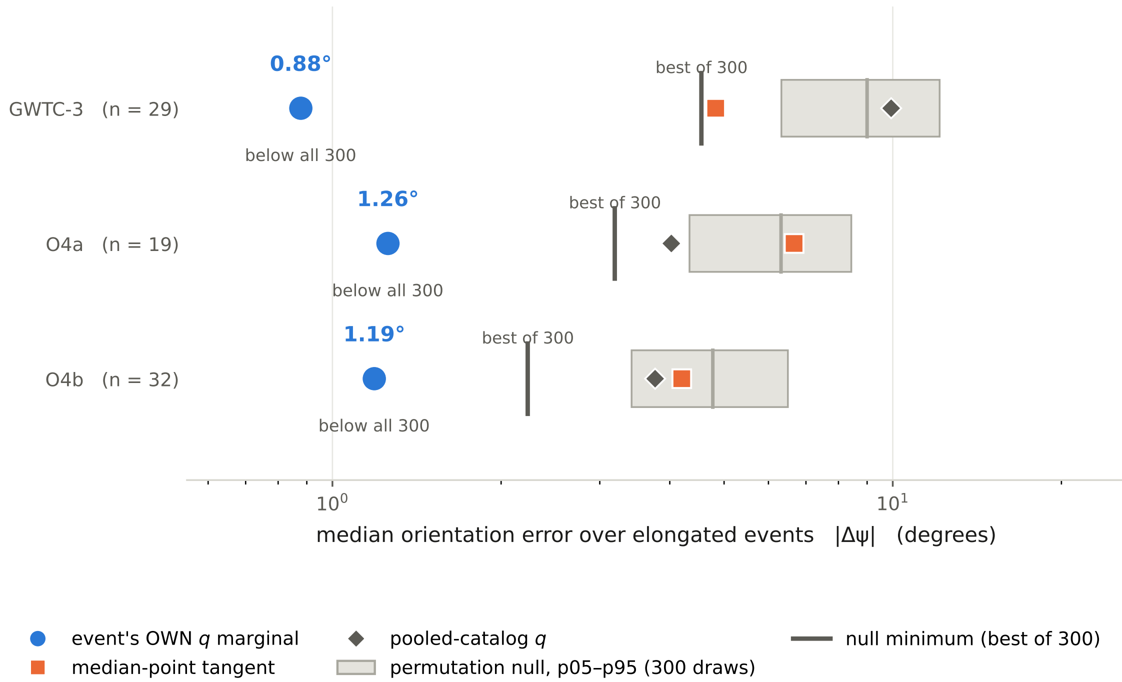

Figure 2: The reconstruction requires the *event's own* mass-ratio marginal. For each catalog the blue circle is the median error using the event's own q marginal; the orange square is the median-point tangent; the grey diamond substitutes a q marginal pooled over the catalog. The grey band is a **permutation null**—the 5th–95th percentile of the catalog-median error when each event is given a *different* event's q marginal, over 300 catalog-stratified draws—and the dark tick marks that null's *minimum*, i.e. the single best of those draws. Own-q lies below the null minimum in every catalog (GWTC-3: own-q 0.88° vs. null minimum 4.55°; O4a: own-q 1.26° vs. null minimum 3.19°; O4b: own-q 1.19° vs. null minimum 2.23°), so the permutation p-value is $< 1/300$ in each. **The band is a null distribution: it is not a confidence interval, not coverage, and not an uncertainty on own-q.**

We describe this as tracking rather than explaining. Both quantities derive from the same posterior and the same curve, so the correlation localises the residual within the geometry without establishing a mechanism, and the corresponding correction does not survive out-of-sample. Replacing the curve by the conditional means — a single Hastie–Stuetzle iteration — reduces the residual in sample, but the improvement is grid-sensitive (median 0.59–1.09° across the sweep) and in the cross-family test it improves both directions while reaching $p < 0.05$ in only one (1.20° → 0.64°, $p = 0.002$; 0.92° → 0.57°, $p = 0.107$). We therefore do not present it as an improved reconstruction.

4.3 Finite posterior thickness

A second contributor is supported out-of-sample. The curve model has zero thickness, whereas a real posterior has finite width perpendicular to the curve — a median of 0.34 of its total spread. Learning that width profile on one waveform family and using it to predict another family's axis improves on the zero-thickness curve in both directions (1.20° → 0.91°, $p = 0.000$; 0.92° → 0.72°, $p = 0.008$).

The *arc-varying* part of that profile is not established: a constant thickness performs as well as the measured profile in one direction, and simple linear or quadratic tapers match or exceed it in both. We therefore attribute no specific fraction of the residual to the measured taper.

5 Robustness

5.1 Frames and coordinates

The measured quantity is a Euclidean principal-axis angle in fixed coordinates and is not reparameterisation invariant. We state that dependence rather than assume it away. Repeating the primary score in detector-frame masses gives 0.32° against 1.07° in the source frame, paired over the same 80 events ($p = 6 \times 10^{-9}$); in $(\log m_1, \log m_2)$ it gives 0.77°. In $(\mathcal{M}_c, q)$ coordinates the construction is degenerate by design, since a constant-$\mathcal{M}_c$ curve is a vertical line and the reconstruction carries no per-event content.

The detector-frame improvement is largely mediated by elongation rather than indicating a better-suited model. Removing the redshift uncertainty that source-frame masses inherit makes posteriors markedly more elongated (median axis ratio 9.8 against 4.6), and at matched axis ratio the two frames no longer separate cleanly: in the 3–5 band the detector frame is in fact worse (5.51° against 1.37°). The source frame remains the locked primary score and the detector-frame comparison is post hoc.

5.2 The elongation threshold

The restriction to elongated posteriors is physical rather than tuned. As the covariance approaches isotropy the principal axis ceases to be defined and the score degrades smoothly toward the random baseline: the median error is 14.6° for axr $\in [1, 1.5)$, then 7.3°, 5.8°, 1.4° and 0.6° for successive bands, approaching but never reaching the 45° expected for a random angle modulo 180°. Nothing distinguishes the chosen threshold of 3: the error falls monotonically and smoothly across thresholds from 1.5 to 10. Figure 3 illustrates the geometry for three events selected by stated rules, including the worst case in the sample.

5.3 Dependence on source parameters

If the geometry is measurement optics, the residual should not depend on source astrophysics. We test this directly. Ranking the 32 elongated O4b events by their curve residual and correlating against nine pre-declared axes — precession χ_p, aligned spin χ_{eff}, mass ratio, redshift, signal-to-noise ratio, chirp mass, total mass, secondary mass, and waveform-family disagreement — with Benjamini–Hochberg control and a partial correlation removing the axis-ratio confound, no physical axis survives. The only surviving correlate is waveform-model disagreement ($\rho = -0.64$, $q = 7 \times 10^{-4}$), a measurement systematic, with a sign indicating that the curve tracks the waveform *ensemble* consensus more closely than any single waveform tracks another.

This is a null result on 32 events and has correspondingly limited power. It bounds the size of any physical dependence; it does not exclude one.

6 Discussion

The orientation of a compact-binary mass posterior is reconstructible from a single one-dimensional marginal of that posterior to about a degree on elongated events, with no coefficient calibrated on the validation catalogs, and the reconstruction holds out-of-sample on two later, disjoint catalogs. It

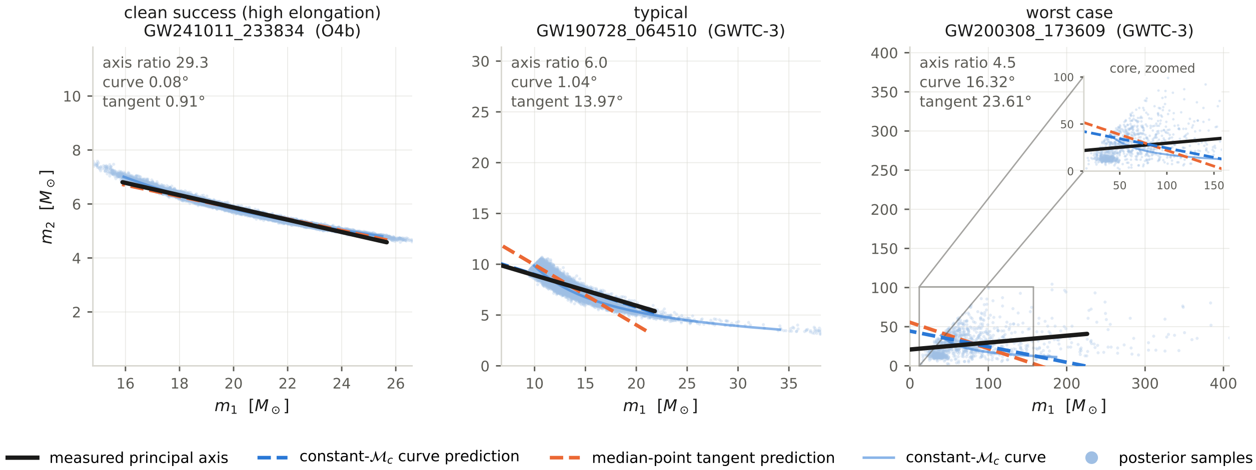

Figure 3: **Illustrative examples** of the posterior geometry and its reconstruction; this panel explains, it does not establish a mechanism. Each panel shows one event's (m_1, m_2) posterior samples, the constant-$\mathcal{M}_c$ curve through the event's median chirp mass, the measured principal axis, and the two predicted orientations. Axes use *equal aspect*: orientation angles are only visually truthful under equal aspect. Events are chosen by deterministic rules recorded in the figure sidecar, not selected by eye: *clean success (high elongation)* GW241011_233834 (O4b, axr 29.3, curve 0.08°, tangent 0.91°); *typical* GW190728_064510 (GWTC-3, axr 6.0, curve 1.04°, tangent 13.97°); *worst case* GW200308_173609 (GWTC-3, axr 4.5, curve 16.32°, tangent 23.61°). The worst case is included by rule so the figure is not a gallery of successes. Across all elongated events the tangent in fact beats the curve for 15% of events, rising to 31% above axr = 8, where the curve's advantage shrinks. No thickness-adjusted axis is drawn: finite posterior thickness is supported out-of-sample but arc-varying thickness is *not* established (Sec. 4). The worst-case panel is a heavy-tailed, high-mass posterior whose dense core occupies a small part of the plotted range, so it carries a zoom inset on that core; the three directions there are the same lines, not a re-fit. That failure is *not* an outlier artifact: trimming to the central 90% of samples leaves the error at 16.2–17.5°, so the curve genuinely misses this event.

is not the trivial observation that a curve outperforms a line: substituting another event's marginal degrades the result several-fold, the achieved error lies below the minimum of 300 permutations, and the reconstruction transfers between separately sampled waveform-family posteriors of the same event. The residual is a systematic of the model, 17.2 times the Monte Carlo resolution of the released samples, to which finite posterior thickness is an out-of-sample contributor.

The tangent baseline quantifies what is being corrected. The principal axis of a Gaussian is the leading eigenvector of its covariance, which for a likelihood expanded quadratically about its peak is the local tangent to the degeneracy; the tangent error is therefore the error incurred by treating the posterior as Gaussian. On elongated events that error is 4.92° against 1.07° once the arc is accounted for — a factor of 4.6. We claim only this elementary geometric fact, and do not invoke the asymptotic statement that posteriors approach a Gaussian with inverse-Fisher covariance, which the argument does not require.

We also do not appeal to the stronger structural language sometimes attached to results of this kind. The "hyperribbon" picture, in which the covariance eigenvalue spectrum spans many orders of magnitude, does not follow from what we measure: in the two-dimensional mass plane the eigenvalue ratio has a median of 3.3, with 28% of events exceeding one order of magnitude and 3% exceeding two. Sloppiness in that sense is a property of the full parameter space, which a two-dimensional marginal cannot establish. Similarly, we do not invoke Čencov's uniqueness theorem for the Fisher–Rao metric: our measured object is a Euclidean angle in fixed coordinates and is demonstrably not invariant (Section 5.1), so an appeal to an invariance theorem would contradict the measurement.

Limitations. (i) This is a posterior-compression result. The mass-ratio marginal is taken from the posterior being reconstructed, so it is not a prediction from source medians or from physics external to the posterior. (ii) It is validated on elongated posteriors; round posteriors have no well-defined orientation and are excluded by construction. Baird et al. [3] note that constant chirp mass ceases to characterise the mass degeneracy as cleanly at higher masses, bounding where such a construction should be expected to hold. (iii) The residual is tracked by the curve's failure of self-consistency, but the corresponding one-step correction is not established out-of-sample, and the arc-varying form of the thickness contribution is not established either. (iv) The measured angle is frame- and coordinate-dependent, and we make no invariance claim. (v) O4a and O4b are disjoint event catalogs, not independent experiments: they share detectors, calibration, waveform families, priors and analysis conventions. (vi) The two out-of-sample scores are not equal evidence. The O4b registration is timestamped in the public record before the data were opened, whereas the O4a registration entered the public repository in the same commit as its results and is attested only by a private history. (vii) Waveform-model choice contributes a per-event orientation scatter of several degrees and is the dominant per-event systematic; on the worst-case event of Figure 3 the error spans 14.4°, 16.9° and 22.7° across its three waveform families. (viii) A first-principles derivation of the curve's width and curvature from the Fisher information remains open.

Not a test of general relativity. The posterior samples analysed here were themselves generated with general-relativistic waveform models. Any structure found in their geometry probes the internal consistency of those inferences, not the theory that produced them, and no bound on modified inspiral phasing follows. Appendix A develops this point quantitatively for a one-parameter generalisation of the chirp-mass exponents, where a naive reading yields an apparent 3.1σ departure from the general-relativistic value that two pre-committed checks demote to 1.5σ. We report it as a diagnostic of the model's own imperfections.

7 Conclusions

The orientation of the component-mass posterior of a compact binary is not merely qualitatively aligned with a constant-chirp-mass contour but quantitatively reconstructible from the event's mass-ratio marginal, to a median 1.26° and 1.22° on two catalogs released after the prediction was fixed. The reconstruction requires the event's own marginal, transfers between waveform families, and leaves a residual that is a property of the model rather than of finite sampling.

Two consequences follow. First, the local Gaussian approximation underlying rapid parameter-estimation tools misstates posterior orientation by a factor of 4.6 more than the arc-corrected construction, which suggests a cheap correction for applications that rely on the orientation. Second, because the geometry is this reproducible, departures from it are measurable, and we find no dependence on any of nine pre-declared source parameters — the only surviving correlate is waveform-model disagreement. A first-principles derivation of the curve's thickness and curvature from the Fisher information, and a strain-level rather than model-level test of the underlying information geometry, remain open.

A A geometric exponent diagnostic

The exponents in Eq. (2) are fixed by the leading-order quadrupole formula. Generalising to a one-parameter family,

$$\mathcal{M}_p = (m_1 m_2)^p (m_1 + m_2)^{1-2p}, \qquad \text{general relativity: } p = 3/5 = 0.600, \tag{3}$$

and fitting p to the posterior orientations isolates the exponent from the mass scale, since orientation is scale-free ($m_1(q; p) \propto q^{-p}(1+q)^{2p-1}$).

On the O4b elongated events we find $p = 0.616 \pm 0.011$; combining O4a and O4b gives $p = 0.628 \pm 0.009$. Taken naively this is a coherent 3.1σ departure from $p = 3/5$. It is a false alarm, and two pre-committed checks demote it. First, the two disjoint catalogs disagree with each other by 0.031, as large as the offset from the general-relativistic value (0.028); this catalog-to-catalog spread is a systematic that the bootstrap statistical error (0.009) does not capture. Including it gives $p = 0.628 \pm 0.009\,(\text{stat}) \pm 0.016\,(\text{syst})$, or 1.5σ. Second, the per-event exponent depends on elongation (Spearman $= -0.41$, coherent in both catalogs), and the most-elongated half yields $p = 0.606$, consistent with the reference value.

The offset is therefore a finite-width imperfection of the leading-order geometric model: coherent across catalogs, and vanishing in the clean limit. We note that coherence alone does not license this conclusion. A systematic common to every event appears in every event, but so would a genuine physical effect, so "coherent, therefore systematic" is not a valid inference. What demotes this particular offset is specific and checkable: the catalog-to-catalog disagreement, and the disappearance of the effect in the most-elongated subset.

B Information anatomy of a detection

The Fisher structure that motivates the posterior orientation also determines when, in frequency, each parameter's information accumulates. Computing the cumulative Fisher information for chirp mass (0PN), mass ratio (1PN) and spin (1.5PN) across 190 O4a and O4b events, the ordering $\mathcal{M}_c < \eta < \chi$ holds universally: chirp mass is determined earliest (~ 35 Hz), spin latest, and the frequency separation between them is a monotonic function of total mass (Spearman -1.00), widest for light, long-inspiral systems and compressed for heavy systems that merge promptly in band.

This is computed from the waveform model rather than measured from strain. The monotonic ordering is therefore a property of the model, and the Spearman value reflects that determinism rather than a statistical result. It is included as context for the measurement-optics picture and supports none of the claims in Sections 3–5. A data-driven version, re-analysing loud events in frequency slices and comparing empirical with predicted information accumulation, is left to future work.

C Reproducibility

All analyses read a single cached extract of the released posterior samples, so that every downstream result shares one deterministic input and can be regenerated without re-reading the parameter-estimation files. The extract performs no subsampling: it stores every sample passing the finiteness and mass-ratio cuts, 23.1 million samples across 972 waveform-group rows, with 972 of those 972 rows held in full. A cache-backed number is therefore a full-sample number and carries no resampling noise.

As a consistency check, an independent pass that re-reads the released files directly, bypassing the cache, returns 1.26° for O4a and 1.19° for O4b, against 1.26° and 1.19° from the cache. For the derivation catalog the two differ by 0.08°, because they resolve one event's preferred waveform group differently; this is a selection convention rather than a sampling effect.

Every number in this paper — in the abstract, the text, Table 1 and every figure caption — is emitted from a committed machine-readable artifact and inserted into the manuscript source as a generated macro rather than typed, so that text, table and figure cannot drift from the analysis or from one another. Figure captions are generated from the same sidecar files that produce the figures.

Data availability

This work is a reanalysis of public parameter-estimation data. It uses no proprietary or embargoed products and generates no new observational data. Three releases are used, obtained from the Gravitational Wave Open Science Center (GWOSC) [11] and Zenodo: GWTC-2.1 [14] and GWTC-3 [15] posterior samples; the GWTC-4.0/O4a release [9], Zenodo record 16053484; and the GWTC-5.0/O4b release [10], Zenodo records 20276106 and 20348006. GWOSC data are distributed under a Creative Commons Attribution licence and are used here under those terms. Analysis code, registered predictions, per-event results and the figure-generation pipeline accompany this work; record numbers and download procedures are given in the accompanying data-availability documentation.

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